

in dynamics of complex systems, yaneer-bar-yam introduces this idea of 'emergent complexity'.

meaning:

read from slide.

where the outcome can't be determined by studying individual parts.

the simplest example of this is one-dimensional cellular automata.

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AFTER PARAMETERS: ->

the parameters are only two: population & day length.

here are 4 interpretations.

interpretation #1.5: born in pairs.

thought:

beings are destined to be with someone that they were born close to (in proximity & age). however, as they live out their lives, they may be close or separated in the world. thought is borrowed from the red thread of fate.

expression:

draw a set of red lines between a being & the person that they're "destined" to be with, over time. the closer they are to each other, the more red lines are drawn, & the stronger they are (in colour and density).

parameters:

population = 4 (shown);

day length = 10;

still image, laid out over time. when all beings of pairs being

visualized die, a still image is generated.

interpretation #2.0: missed connections.

thought:

we walk by so many people. when they are within a certain distance, we have a short window of time to connect with each other in physical-space, until we are distant again.

expression:

draw a line from each being to other beings around them within a specific radius, with the distance between them signifying the intensity of a possible connection.

parameters:

population: 1000,
day_length: 10,

notice that the connections don't disappear, but fade as beings move away from each other.

interpretation #3.0: all the world's a mesh.

thought:

all beings are connected to each other via an invisible mesh.

expression:

treat the position of each being as a vertex. generate a triangular-mesh by finding tuples close to each other, so that no point is inside a face. keep generating the mesh over time, as beings move.

parameters:

population: 1000
day-length: 5

images of interpretation-3 at different points in time.

interpretation #4.1: meetings are ripples.

thought:

meeting people can be resounding. when two beings meet, a ripple is sent through space & time; that may affect other beings.

expression:

connect beings close to each other with a black line. when two beings collide, send out a circular ripple expanding from the point of collision. the closer the beings are, the further the ripple goes. lay this out over time.

parameters:

population: 500

day-length: 10